Multinomial Logistic Regression

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Nominal Target Variable

It is easy to generalize the binary logistic regression model to the ordinal logistic regression model. However, we need to change the underlying model and math slightly to extend to nominal target variables.

In binary logistic regression we are calculating the probability that an observation has an event. In ordinal logistic regression we are calculating the probability that an observation has at most that event in an ordered list of outcomes. In nominal (or multinomial) logistic regression we are calculating the probability that an observation has a specific event in an unordered list of events.

We will be using the alligator data set to model the association between various factors and alligator food choices - fish, invertebrate, birds, reptiles, and other. The variables in the data set are the following:

Variable	Description
size	small (< 2.3 meters) or large (> 2.3 meters)
lake	lake captured (George, Hancock, Oklawaha, Trafford)
gender	male or female
count	number of observations with characteristics

Generalized Logit Model

Instead of modeling the typical logit from binary or ordinal logistic regression, we will model the generalized logits. These generalized logits are built off target variables with m categories. The first logit will summarize the probability of the first category compared to the probability of the reference category. The second logit will summarize the probability of the second category compared to the probability of the reference. This continues $m-1$ times as the last logit compares the $m-1$ category to the reference category. In essence, with a target variable with m categories, we are building $m-1$ logistic regressions. However, unlike the ordinal logistic regression where we have different intercepts with the same slope parameters, in nominal logistic regression we have both different intercepts and different slope parameters for each model.

The logits are also not traditional logits. Instead of the natural log of the odds, they are natural logs of relative risks - ratios of two probabilities that do not sum to 1.

Let's see how to build these generalized logit models in each of our softwares!

R Python SAS

The data set is structured slightly differently than previous data sets. Let's examine the first 10 observations with the `head` function and the `n = 10` option.

▼ Code

```
head(gator, n = 10)
```

```
            size         food    lake gender count
1  <= 2.3 meters         Fish Hancock   Male     7
2  <= 2.3 meters Invertebrate Hancock   Male     1
3  <= 2.3 meters        Other Hancock   Male     5
4   > 2.3 meters         Fish Hancock   Male     4
5   > 2.3 meters         Bird Hancock   Male     1
6   > 2.3 meters        Other Hancock   Male     2
7  <= 2.3 meters         Fish Hancock Female    16
8  <= 2.3 meters Invertebrate Hancock Female     3
9  <= 2.3 meters      Reptile Hancock Female     2
10 <= 2.3 meters         Bird Hancock Female     2
```

Notice the *count* variable. In the first observation we can see that we have seven alligators that are small, males captured on Lake Hancock and eat fish. The same applies for each row in our data set. Even with only 56 rows in our data set, we have 219 observations based on the counts.

To account for the multiple counts in a single row of our data set we use the `weight =` option in the `vglm` function (like we saw with ordinal logistic regression) with the *count* variable. The formula structure for the `vglm` function is the same as with previous functions in logistic regression. The target variable *food* needs to be turned into a factor with the `factor` function. Instead of `cumulative` for the `family =` option like we had in ordinal logistic regression, we use the `family = multinomial` option. We set the reference level specified with the `refLevel` option so we know what R chooses. By default the `vglm` function will use the last level of the variable. Here, our reference level is fish because it has the largest proportion of our observations. Similar to before, the `summary` function will display our needed results.

▼ Code

```
gator$food <- factor(gator$food)

library(VGAM)

glogit.model <- vglm(food ~ size + lake + gender,
                     data = gator,
                     family = multinomial(refLevel = "Fish"),
                     weight = count)
summary(glogit.model)
```

```
Call:
vglm(formula = food ~ size + lake + gender, family = multinomial
    data = gator, weights = count)

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept):1        -2.43211    0.77066      NA       NA
(Intercept):2         0.16902    0.37875   0.446  0.65541
(Intercept):3        -1.43073    0.53809  -2.659  0.00784 **
(Intercept):4        -3.41604    1.08513      NA       NA
size> 2.3 meters:1    0.73024    0.65228   1.120  0.26292
size> 2.3 meters:2   -1.33626    0.41119  -3.250  0.00116 **
size> 2.3 meters:3   -0.29058    0.45993  -0.632  0.52752
size> 2.3 meters:4    0.55704    0.64661   0.861  0.38898
lakeHancock:1         0.57527    0.79522   0.723  0.46943
lakeHancock:2        -1.78051    0.62321  -2.857  0.00428 **
lakeHancock:3         0.76658    0.56855   1.348  0.17756
lakeHancock:4         1.12946    1.19280   0.947  0.34369
lakeOklawaha:1       -0.55035    1.20987  -0.455  0.64919
lakeOklawaha:2        0.91318    0.47612   1.918  0.05511 .
lakeOklawaha:3        0.02606    0.77777   0.034  0.97327
lakeOklawaha:4        2.53026    1.12212   2.255  0.02414 *
lakeTrafford:1        1.23699    0.86610   1.428  0.15323
lakeTrafford:2        1.15582    0.49279   2.345  0.01900 *
lakeTrafford:3        1.55776    0.62567   2.490  0.01278 *
lakeTrafford:4        3.06105    1.12973   2.710  0.00674 **
genderMale:1         -0.60643    0.68885  -0.880  0.37867
genderMale:2         -0.46296    0.39552  -1.171  0.24180
genderMale:3         -0.25257    0.46635  -0.542  0.58810
genderMale:4         -0.62756    0.68528  -0.916  0.35978
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,2]), log(mu[,3]/mu[,2
log(mu[,4]/mu[,2]), log(mu[,5]/mu[,2])

Residual deviance: 537.8655 on 200 degrees of freedom

Log-likelihood: -268.9327 on 200 degrees of freedom

Number of Fisher scoring iterations: 6

Warning: Hauck-Donner effect detected in the following estimate(
'(Intercept):1', '(Intercept):4'


Reference group is level  2  of the response
```

In the output above we see similar things to the output we have seen previously. R summarizes the parameter estimates, their standard errors, and corresponding p-values. The main difference in the parameter estimates is the multiple intercepts and slope coefficients for each variable. Since we have five levels of our target variable we have four logistic regression models and therefore four intercepts and four slope parameters for each variable.

Interpretation

Similar to binary and ordinal logistic regression models, we exponentiate the coefficients in our nominal logistic regression model to make them interpretable. However, the interpretation changes since these are not odds ratios anymore, but relative risk ratios. Let's use the coefficient for the birds model and the *size* variable. It would be **incorrect** to say that the probability of eating birds is 2.076 times as likely for large alligators compared to small alligators. The **correct** interpretation would be that the predicted **relative probability** of eating birds **rather than fish** is 2.076 times as likely in large alligators compared to small alligators. Notice how our interpretation is in comparison to the reference level. Sometimes these are called **conditional** interpretations.

Let's see how to get these from each of our softwares!

R Python SAS

As with previous models, we just need to use the `exp` function to exponentiate the coefficients from our nominal logistic regression model to get the relative risk ratios.

▼ Code

```
100*(exp(coef(glogit.model))-1)
```

```
     (Intercept):1      (Intercept):2      (Intercept):3      (I
        -91.214894          18.414855         -76.086613
size> 2.3 meters:1 size> 2.3 meters:2 size> 2.3 meters:3 size> 2
        107.557742         -73.717347         -25.217238
     lakeHancock:1      lakeHancock:2      lakeHancock:3      la
         77.760394         -83.144823         115.238205
    lakeOklawaha:1     lakeOklawaha:2     lakeOklawaha:3     lak
        -42.325256         149.223994           2.640013
    lakeTrafford:1     lakeTrafford:2     lakeTrafford:3     lak
        244.522908         217.663308         374.818559
      genderMale:1       genderMale:2       genderMale:3       g
        -45.470516         -37.058401         -22.319776
```

Predictions & Diagnostics

Nominal logistic regression has a lot of similarities to binary logistic regression:

- Multicollinearity still exists
- Non-convergence problems still exists
- AIC and BIC metrics can still be calculated
- Generalized R^2 remains the same

There are some inherent differences though between binary and nominal logistic regression. A lot of the diagnostics cannot be calculated for multinomial logistic regression. ROC curves and residuals cannot typically be calculated because there is actually more than one logistic regression occurring.

Predicted probabilities are actually for each category though. Let's see how we can get predicted probabilities from each of our softwares!

R Python SAS

In R we have the same process of getting predicted probabilities in multinomial logistic regression as we did for binary and ordinal logistic regression. The `predict` function will score any data set. The `type = response` option is used to get individual predicted probabilities for each category.

▼ Code

```
pred_probs <- predict(glogit.model, newdata = gator, type = "re
head(pred_probs)
```

```
       Bird      Fish Invertebrate     Other    Reptile
1 0.0511489 0.6006519   0.07545711 0.2401562 0.03258584
2 0.0511489 0.6006519   0.07545711 0.2401562 0.03258584
3 0.0511489 0.6006519   0.07545711 0.2401562 0.03258584
4 0.1102286 0.6236514   0.02059152 0.1864723 0.05905622
5 0.1102286 0.6236514   0.02059152 0.1864723 0.05905622
6 0.1102286 0.6236514   0.02059152 0.1864723 0.05905622
```

The five variables are the individual predicted probabilities for each of the categories of our target variable. Unlike binary logistic regression where we typically did not use a cut-off of 0.5 to determine which category we predict, in nominal logistic regression we typically pick the category with the highest probability as our predicted category.

Accuracy is typically not as high in logistic regressions with more than two categories because there are more chances for incorrect predictions.

← Ordinal Logistic Regression